

Career Compass — Product Audit & Feature Recommendations

Companion to `IDEA_ANALYSIS.md` (market/business view). This document is a **feature-by-feature product audit** of the actual codebase, with explicit **KEEP / CHANGE / ADD / DELETE** recommendations and severity flags.

Severity legend: ● P0 (ship-blocker / security / data-integrity) · ● P1 (next 30 days) · ● P2 (next 90 days) · ● Nice-to-have

0. Top-Line Findings (read this first)

1. ● **P0 — Anthropic API key is shipped to the browser.** `src/pages/Results.tsx:216–247` reads `import.meta.env.VITE_ANTHROPIC_API_KEY` and posts directly to `https://api.anthropic.com/v1/messages` with the header `anthropic-dangerous-direct-browser-access: true`. Vite bakes any `VITE_`-prefixed variable into the client bundle, meaning the key is visible to every user via DevTools. **Anyone can scrape it and run arbitrary Anthropic API spend on your account.** Move this to a Supabase Edge Function or a Netlify Function and call it from the browser.
2. ● **P0 — Question counts are inconsistent across the UI.** Marketing copy (`Index.tsx`) promises *"15 + 15 + 5"*. Dashboard (`Dashboard.tsx:104–123`) says *"20 questions / 24 questions / 12 questions"*. Results page (`Results.tsx:353–355`) hard-codes `minComplete = 20 / 24 / 12` in `findBestAssessmentId`. The marketing page lies to users about the test length (8/10/5 min totals advertised; real time is ~25–30 min). Pick one number set and align them.
3. ● **P0 — Debug console.log s ship to production.** `Results.tsx:366–387` logs the entire user answer map (`aptMap` , `intMap` , `perMap`) plus assessment IDs and calculated scores. Same in `AssessmentEngine.tsx:171, 234, 256`. This is a privacy leak (PII goes to browser console + any analytics scraper) and looks unprofessional in a product targeting parents. Wrap in `if (import.meta.env.DEV)` or remove.
4. ● **P1 — Two sources of truth for careers, with different data.** `src/lib/scoring.ts` has a `topCareers` map (5 careers/stream, title + description). `src/pages/Careers.tsx` has its own `ALL_CAREERS` array (18 careers, with `skills` , `salary` , `industry`). They overlap but don't reference each other. When you change "Data Scientist" salary you have to edit two files — and the Results page top-5 will eventually drift from the public careers catalogue.
5. ● **P1 — Results page and Careers page are disconnected.** A student gets a Science recommendation, sees 5 career chips on Results, then visits Careers and has to manually filter to "Science". There's no *"See all Science careers →"* link from Results → Careers, and no *"You were recommended Science"* banner on Careers. This is one CTA away from being a much better funnel.
6. ● **P2 — Stream model is hard-coded in a SQL CHECK constraint.** `results.stream CHECK (stream in ('Science', 'Commerce', 'Humanities'))` makes it expensive to add a 4th stream (Vocational, Design, Liberal Arts) when NEP 2020 / state boards move that way. Loosen to a `text` column + a `streams` reference table.

1. Page-by-Page Audit

1.1 `Index.tsx` (Landing) — Keep, polish

What works:

- Clean hero, three "How it works" cards (Aptitude / Interest / Personality), feature grid, parent-share callout, dual CTA (Start Free / Explore Careers).
- India-context badge (*"AI-Powered Career Guidance for Indian Students"*).
- Stat row (3 Tests · ~25 min · 100% Free) — concrete, builds trust.

What's broken / weak:

- ✗ The "How it works" question counts (15/15/5 = 35 total) **don't match** what the user actually faces (20/24/12 = 56 total). Either the marketing under-promises or the Dashboard over-promises — they cannot both be right.
- ✗ "Join thousands of Indian students who've found their ideal stream" — there's no evidence of "thousands" anywhere. Either prove it (counter from `results` table) or soften the claim. Right now it's a credibility hole the first counsellor or journalist will pick at.
- ✗ No school logos, no testimonial, no sample report preview. Indian parents need at least *one* social-proof anchor to convert.
- ✗ The "AI-Powered Analysis" feature card overpromises — the AI is one 3–4 sentence message, not analysis. Reframe as *"Personal Counsellor Note"*.

Recommendations:

Action	Priority	Note
CHANGE the test-count copy to match reality	● P0	Single source of truth.
ADD a real "X assessments completed this week" counter	● P1	Pull from <code>select count(*) from results where created_at > now() - '7 days'::interval</code> . Honest social proof.
ADD a "See sample report" link → static PDF	● P1	Removes the "what do I actually get?" objection before signup.
ADD Hindi toggle (and one more lang)	● P2	Tier-2/3 market unlock.
DELETE "thousands of students" unless verifiable	● P0	False social proof is a legal/PR risk.

1.2 Dashboard.tsx (Student Hub) — Strong, but lying about test length

What works:

- Clean progress bar (X/3 assessments), per-card status (`Start` / `Continue` / `View Responses`), locked Results card with a clear unlock condition.
- Attempt History** is genuinely impressive for an MVP — shows newest-first, "Latest" badge, "Same as prev." badge when retake produces identical scores, mini score bars per stream, deduplication logic for orphan retake rows. This is the kind of thing incumbents don't do.
- `StreamPreferenceCard` shown only until the student picks — good progressive disclosure.

What's broken / weak:

- ✗ Test-length copy (lines 104–123): `"20 questions · ~15 min"` , `"24 questions · ~12 min"` , `"12 questions · ~8 min"` — totals 56 Q / ~35 min. Marketing page said 35 Q / 23 min. **This is the most-touched copy in the app — fix the inconsistency first.**
- ✗ "Same as prev." badge logic compares only the final scores, not the answer-set — so a student who actually changed answers but happened to land on the same scores gets told they answered similarly. Misleading.
- ✗ Attempt History card "Retake all assessments to generate a new result." This text appears on *every* non-latest attempt — but the action (Retake) is only available from the Results page. Dead-end copy.
- ✗ No empty-state for the very first visit — student sees a "0/3" progress bar and three cards. A 1-sentence "Take your first assessment to begin →" with an arrow to the Aptitude card would help.

Recommendations:

Action	Priority	Note
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CHANGE test counts/durations to a single source	● P0	Put them in <code>src/lib/config.ts</code> and import.
CHANGE "Same as prev." to compare answer-vectors, not scores	● P1	Easy: hash the answer map.
ADD "Continue where you left off" jump-card on top when any test is in progress	● P1	Reduces re-entry friction.
ADD a small "time spent so far" stat per card (we have <code>started_at</code>)	●	
DELETE the "Retake all assessments to generate a new result." footer on history cards	● P1	Or wire it as an actual Retake link.

1.3 AssessmentEngine.tsx (the 3-test workhorse) — Solid; clean up debug

What works:

- Resume-from-pause is real: `pause_position`, restore responses, jump to first unanswered.
- Single delete-then-insert auto-save per answer (avoids unique-constraint conflicts on `(assessment_id, question_id)`).
- Bulk re-save of all answers before marking `status='completed'` — won't mark complete on partial save failure.
- Three answer-types in one component (`mcq / likert / options`) — clean abstraction.
- "Read-only review" mode after completion with green/red highlighting on aptitude — students get to see what they got right. Nice touch.

What's broken / weak:

- ❌ **Six `console.log` statements** in the hot path (lines 148, 171, 234, 256, plus ❌ and ✅ emoji-debug). Strip before any pilot.
- ❌ Submit button when not all answered says `Submit (X/N)` with orange colour and `AlertCircle` icon — but pressing it *navigates* to the first unanswered, doesn't submit. The label "Submit" is misleading. Should say `Go to Q{unanswered}` or render a different button.
- ❌ Likert emoji mapping: `value '1' → 😐`. A student told "Not at all interested" sees a neutral face; "1 = strongly disagree" usually maps to 😞 or 🙄. Minor, but Indian users sometimes interpret 😐 as "indifferent" not "no".
- ❌ No timer. Aptitude questions like "If a train travels 60 km in 45 minutes..." are *trivially Google-able* when the student has 15 minutes per question. A 60-90 second per-question soft timer (with skip-with-warning) would dramatically improve validity.
- ❌ No "back button protection" — students can hit browser-back mid-question and may lose their place (state is server-backed but UX is rough).
- ❌ No support for question types beyond MCQ/Likert/options — no slider, no rank-order, no scenario-based. Limits the psychometric headroom you can ever reach.
- ❌ Aptitude questions have a single deterministic `correctAnswer` — no support for partial credit, no item difficulty weighting, no IRT.

Recommendations:

Action	Priority	Note
DELETE all <code>console.log/console.warn</code> in this file before pilot	● P0	
CHANGE "Submit (X/N)" → "Review unanswered (X left)" when incomplete	● P1	Button label honest.
ADD an optional per-question 90-second soft timer (off by default; on for schools)	● P2	Big validity unlock.
ADD "Skip & come back" explicit action (today the dot-nav is the only way; not discoverable)	● P1	

ADD anti-cheat: shuffle option order per session	● P2	Cheap; meaningful for retakes.
DELETE the Likert 😊 emoji for "1" — replace with 🙄 or just label	●	

1.4 Results.tsx (the payoff screen) — Powerful, security-leaky, debug-noisy

What works:

- Stream hero with emoji + recommendation + score-bar breakdown + delta arrows vs. previous attempt (⬆5 / ⬆3) is genuinely well-designed. The delta is the kind of detail that earns parent trust.
- "Stream Preference Alignment" panel (✅ "Your instincts were right" / 🔍 "Your results suggest a different path") is a *brilliant* affective design choice for the Indian context where students/parents pre-commit to a stream. Keep this.
- Competitive Exam Roadmap (5 exams/stream, with level and difficulty badges) is the standout feature that differentiates Career Compass from a generic quiz. Schools will want this.
- Retake confirmation modal is appropriately scary; preserves history.
- "Why this stream?" callout with reasoning text.
- StrictMode double-effect guard via loadResultsRunning ref — careful engineering.

What's broken / weak:

- ● **The Anthropic API key leak** (lines 216–247). Severity: high. Mitigation: move to server function.
- ● **9 console.logs** (lines 366–387) that dump user answers + computed scores. Strip.
- ✗ **Score insertion has a triple fallback** (insert with attempt_number → insert without → another insert with different shape via PGRST06 fallback). The code comment cites a stale Supabase schema cache. This is firefighting around a migration that should just be properly applied; the data model now has 3 ways an attempt row can look, and the dashboard dedup logic has to clean it up. **Pay this debt** — run a one-time backfill and remove the fallbacks.
- ✗ Grade-for-scoring is **detected from question-ID prefixes** (apt_d_ , apt_c_) at line 376–378. Magic strings. If someone renames a question, scoring silently mis-attributes the grade. Store grade_group on the assessments row at creation time.
- ✗ getStreamRecommendation() is called **twice** — once during loadResults (line 388) and again on render at line 620 — using different inputs. Once would suffice, and using the cached result avoids re-computing for a stored row.
- ✗ "Why this stream?" reasoning is **the same 3-paragraph hard-coded text for every student** with the same stream (scoring.ts:126-130). The whole point of the assessment is personalisation; the most-read text on the results page is the least personalised. The AI narrative is meant to fix this but is optional and rendered separately.
- ✗ The AI narrative builder *passes summaries, not numbers* to the model — good for hallucination control, but the result is necessarily generic ("you answered X questions"). It can't say *"you scored 85 in numerical and 35 in verbal"* even if you want it to.
- ✗ The PDF download (generatePDF.ts , 269 LOC) is client-side via jsPDF — Indian-context fonts (Hindi/Tamil names) will silently break. Test with non-ASCII names.
- ✗ The Share button copies the link but **doesn't open WhatsApp**. The single most-likely sharing surface in India is one tap away and you're using clipboard — a regression.
- ✗ The exam roadmap is hard-coded in this 926-line file (lines 22–174). Extract to src/lib/examRoadmap.ts or, better, a Supabase table so it can be updated without a deploy.

Recommendations:

Action	Priority	Note
CHANGE AI narrative call → Supabase Edge Function	● P0	Stop the key leak.
DELETE all debug console.logs	● P0	
CHANGE Share button → "Share on WhatsApp" with wa.me/?text=... deep link	● P1	5-line change, huge funnel uplift.

CHANGE reasoning from hard-coded text → generated from actual sub-scores	🟡 P1	"Your numerical (85) and spatial (78) scores are well above average for Science aspirants."
CHANGE grade detection → store grade_group on assessment row	🟡 P1	Removes magic-string fragility.
CHANGE insert flow → single canonical insert (run a select 1 to warm schema cache, fix migration)	🟡 P1	Pay the tech debt.
ADD "See all 6 Science careers →" link to /careers?stream=Science	🟡 P1	Cross-page link Results → Careers.
ADD download-the-PDF preview thumbnail before downloading	🟢	
EXTRACT EXAM_ROADMAP to its own module or table	🟡 P2	

1.5 Careers.tsx (Career Library) — Standalone island; needs integration

What works:

- 18 careers across 3 streams, with skills , salary (₹ range), industry .
- Stream filter pills + free-text search across title/industry/description. Snappy on small lists.

What's broken / weak:

- ❌ **Two careers data sources** that disagree. Careers.tsx lists "Aerospace Engineer ₹6L-25L"; scoring.ts doesn't even have it in the Science top-5. The PDF report uses one list; the public library uses another. Pick one.
- ❌ No deep-link by stream (?stream=Science) — the React state defaults to 'All' , so a Results → Careers link can't pre-filter.
- ❌ No deep-link by career (/careers/data-scientist) — careers are not routable individually, so you can't share a single career page, can't SEO it, can't Google-rank it. **For a free top-of-funnel product, this is the single biggest organic-growth miss.** Each career page is a potential SEO asset ("data scientist career in India" — high-intent query).
- ❌ Salary ranges are India-rough but no city granularity (Bangalore vs Patna), no source citation, and "Unlimited potential" for Entrepreneur is funny but not useful.
- ❌ No "Day in the life", no "Required degree path", no "Top colleges", no "Coaching/exam to crack" — i.e. the page tells me the job exists but not how to get it. After-result moment is exactly when the parent wants this depth.
- ❌ Search has no debounce; tiny list so fine today, but breaks at 200 careers.

Recommendations:

Action	Priority	Note
CONSOLIDATE careers into one source of truth (src/data/careers.ts or careers table)	🟡 P1	Single change to add Aerospace Engineer once.
ADD ?stream= query param + auto-select	🔴 P0	Wires Results → Careers.
ADD individual career routes /careers/:slug	🟡 P1	Each becomes an SEO asset.
ADD career-page sections: Day-in-the-life · Path-to-this-career · Top colleges · Required exams · Famous Indians in this role	🟡 P2	This is the page parents share on WhatsApp.
DELETE "Unlimited potential" salary for Entrepreneur → realistic range or specific cohort data	🟡 P2	
ADD "Save for later" / "Add to my shortlist"	🟢	Drives return visits.

1.6 Signup.tsx / ParentSignup.tsx / CounsellorSignup.tsx — Working, no progressive profiling

What works:

- Three role-specific signup flows with appropriate fields (counsellors collect `school_name`).
- Supabase `handle_new_user` trigger auto-creates the `profiles` row server-side — robust.
- Forgot-password flow exists.

What's broken / weak:

- **✗ One signup form, all fields required up front** for the student (Phone, Grade, School, City). High abandonment risk for a free-trial product. The optimal pattern is: signup with `email + password + full_name` only → progressive profile during the assessment (Grade asked on Aptitude page, City asked on Results page).
- **✗ No social signup (Google)** — adding 1 line via Supabase doubles signup conversion in most consumer flows. Indian students/parents largely use Gmail.
- **✗ Phone field stored but never used** (no OTP, no WhatsApp). Either use it or remove it.
- **✗ Counsellor signup gates them straight into a counsellor dashboard with no school-verification.** A coaching brand can pretend to be a school counsellor and harvest student emails. Add email-domain verification or admin approval.
- **✗ Parent-link by share-token UX is unclear from the code** — needs to be a one-step "Paste the link your child shared" flow on the parent dashboard.

Recommendations:

Action	Priority	Note
CHANGE signup → email/name/password only; ask grade/school later	● P1	Conversion lift.
ADD Google OAuth	● P1	Supabase 1-liner.
ADD counsellor email-domain check or admin approval	● P1	Trust + safety.
DELETE phone field if unused	● P2	Or wire it to WhatsApp result delivery (much better).

1.7 ParentDashboard.tsx — Functional, but the parent surface is the weakest of the four roles

(From `1s`-sized 265 LOC + the data model.)

What works:

- Parents can sign up with a `parent` role and see linked students via `parent_student_links`.
- Public share-token route (`/share/:token`) lets parents view a child's result *without an account* — keep this; it's the viral mechanic.

What's broken / weak (inferred from surface area):

- **✗ The whole parent journey is *read-only*.** Parents can't ask follow-up questions, can't compare their child against the cohort, can't book a counsellor call, can't get a "what does this mean for my child" walkthrough.
- **✗ No multi-child support is obvious** (parents in India often have 2 children in different grades).
- **✗ No "next steps for you" panel** (parents leave the report without knowing what to do this week).
- **✗ Almost no parent-specific copy:** the report is written *to the student*, with the parent peering over their shoulder. Parents need their own narrative.

Recommendations:

Action	Priority	Note
ADD "Parent's Guide to This Result" — a parallel 4-paragraph narrative addressed to the parent	● P1	Highest-value content add.
ADD "Ask the AI counsellor" 5-message chat scoped to <i>this child's</i> result	● P1	Parent-paid premium tier.
ADD "Compare with other 10th-graders in your city" anonymised cohort view	● P2	Becomes more useful at scale.
ADD "Book a 15-min counsellor call" CTA (initially routed to a Calendly)	● P1	Monetisation surface.
ADD multi-child support (one parent → many parent_student_links)	● P2	Schema already supports it; UX doesn't.

1.8 CounsellorDashboard.tsx (499 LOC) — The B2B2S linchpin

What works:

- Per-school roster, can see each student's status.
- Migration history (005-008) shows this surface was iterated on hard — including dropping infinite-recursion RLS policies in 008.

What's broken / weak:

- ❌ **school_name is free-text** (profiles.school_name TEXT). Two students at "DPS RK Puram" and "D.P.S R.K Puram" end up in different counsellor groupings. Needs a schools reference table with city + ID.
- ❌ Migration 008 *dropped* the counsellor-can-see-students RLS policies because of recursion. Either those policies were rewritten via SECURITY DEFINER (like admin in migration 009) — confirm this — or counsellor access is currently broader/narrower than intended.
- ❌ No bulk-import of students (CSV upload by school). Today a school onboarding means each student signs themselves up and types the school name correctly — that won't scale.
- ❌ No exportable school report ("here are your 200 students' streams broken down"). Principals will ask for this on day 1.
- ❌ No counsellor messaging surface — can't message a student or parent.

Recommendations:

Action	Priority	Note
ADD schools reference table with controlled list (seed with CBSE/ICSE school directory)	● P1	Single biggest data-quality win.
CHANGE counsellor RLS → SECURITY DEFINER function pattern (mirror is_admin())	● P0	Audit + verify current access; the migration history suggests this may be broken.
ADD CSV bulk-import for students	● P1	First school-paid feature.
ADD School-level report (PDF + dashboard) showing stream distribution, completion rate, gender split	● P1	Principal asks for it.
ADD Counsellor → student/parent in-app message	● P2	Eventually a chat surface.

1.9 AdminDashboard.tsx + Migration 009 — Good admin foundation

What works:

- `is_admin()` SECURITY DEFINER helper is the *correct* pattern (sidesteps RLS recursion).
- 281 LOC dashboard exists.

What's broken / weak:

- ✗ No audit log of admin actions.
- ✗ Promoting a user to admin is done via raw SQL (`UPDATE public.profiles SET role = 'admin'...`). Fine for now, but should become a 2-admin approval flow before scale.

Recommendations:

Action	Priority	Note
ADD <code>admin_audit_log</code> table (who, when, what they viewed/changed)	🟡 P2	
ADD Admin-only Anthropic usage dashboard	🟡 P2	Will matter once you have spend.

1.10 Cross-cutting surfaces — Missing

Missing surface	Why it matters	Severity
Analytics / event tracking	No PostHog, no GA. You don't know your assessment-completion rate, drop-off question, time-on-page. Flying blind.	🟡 P1
Error monitoring	No Sentry. Production errors will only be surfaced by user complaints.	🟡 P1
Automated tests	No <code>*.test.*</code> , no Vitest config, no Playwright. Scoring algorithm is the most-critical 50 lines in the codebase and has zero tests.	🟡 P1 — add unit tests for <code>calculateScores</code> first.
i18n	All copy is English-only. Hindi alone unlocks ~50% more of the SAM.	🟡 P2
Accessibility	No <code>aria-label</code> on the icon-only buttons (Pause, Share, Download), no skip-link, no focus-ring testing. ~10% of students have some accessibility need; some schools require A11Y compliance to procure.	🟡 P2
Mobile keyboard handling	Long-form assessment on a phone: nothing scrolls the question into view when the keyboard opens for "Other" inputs. Not a problem today since no free-text answers exist — but the moment you add a "What career did you have in mind?" field, this will bite.	🟢
Offline / poor-connectivity mode	India has flaky 3G/4G in tier-3. Today every answer triggers a delete+insert round trip. Buffer answers locally + flush in batch.	🟡 P2
SEO	No meta tags beyond defaults, no sitemap, no per-career pages. Free careers content is the lowest-CAC channel and is not being captured.	🟡 P1

2. Cross-Feature: What to ADD that isn't here yet

In rough priority order:

1. ● **Stream-comparison view** — "Your Science score (78) is 12 points ahead of Commerce (66) and 18 ahead of Humanities (60). Here's why." A radar chart + a delta table replaces the generic *"Why Science?"* paragraph. *(Build effort: 1 day.)*
2. ● **Parent narrative + Parent chat** — Already covered above. The single biggest monetisable feature you don't have. *(Build effort: ~1 week, including server-side AI route.)*

3. ● **Subject-combination recommendation (Class 11 picks)** — Today you recommend a stream. Indian students don't choose a stream — they choose a *combination* (PCM, PCB, PCMB, Commerce-with-Maths, Commerce-without-Maths, Humanities + Psych/Eco/PolSci, etc.). This is the most-asked follow-up question and a clear depth upgrade. (*Build effort: 3-5 days; mostly content.*)
4. ● **"Compare with peers" cohort percentile** — "Your numerical score is in the 78th percentile for Class 10 students in Karnataka." Requires $N \geq 500$ to be honest; trivial to compute once you have data. Massive psychological pull for parents. (*Build effort: 1 day at scale.*)
5. ● **WhatsApp result share** — Replace clipboard-only Share with a `wa.me/?text=` deep link. (*Build effort: 30 minutes.*)
6. ● **Adaptive item-bank with IRT-lite** — Don't give every student the same 15 aptitude questions. Bank of 60; serve 15 based on previous answers (item difficulty calibrated from response patterns). Doubles validity at the same test length. (*Build effort: 2-3 weeks + a psychometrician.*)
7. ● **Counsellor-led group session mode** — A counsellor in a classroom of 40 wants a "presentation mode" where they can walk students through a sample report on screen. Currently they have to share their own logged-in dashboard, which is messy. (*Build effort: 3-5 days.*)
8. ● **College + course shortlist post-Class-12** — Stream → degree → top colleges accepting that degree → CUET subject mapping. This is the *natural* upsell to the existing Class-10 product, and it's where iDreamCareer makes most of its revenue. (*Build effort: ~1 month — content-heavy.*)
9. ● **Gamification of retakes** — Today retaking 3 weeks later shows a "Same as prev." badge with no encouragement. Make it a growth ritual ("How have I changed?" once per quarter). Email reminder + diff view.
10. ● **Counsellor → Parent intro on counsellor-completion** — When the counsellor closes the student's session, auto-trigger a "Your school counsellor recommends a 15-min parent debrief" email.

3. What to DELETE / SIMPLIFY

Thing	Why
Inline <code>EXAM_ROADMAP</code> constant in <code>Results.tsx</code> (lines 22-174)	Bloats the page file to 926 LOC. Move to a module or a Supabase table.
Triple-fallback <code>INSERT</code> in <code>Results.tsx</code> <code>loadResults</code> (lines 413-431)	Fix the underlying schema-cache issue, kill the fallback chain.
<code>STREAM_COLORS</code> duplicated in <code>Results.tsx</code> , <code>Careers.tsx</code> , <code>Dashboard.tsx</code>	Centralise in <code>src/lib/streamTheme.ts</code> .
<code>topCareers</code> in <code>scoring.ts</code> VS. <code>ALL_CAREERS</code> in <code>Careers.tsx</code>	Pick one.
All 14 <code>console.log</code> / <code>console.warn</code> calls	Strip or wrap in <code>if (import.meta.env.DEV)</code> .
<code>phone</code> field on profiles (if unused)	Either use it for OTP/WhatsApp or delete.
Phase D counsellor dashboard "removed from user journey" (commit 973b6c8)	Either fully removed or fully wired. Half-removed features are the worst kind of tech debt. Audit and finish the job.
<code>debug</code> columns / logging in <code>Results.tsx</code>	See P0 above.

4. The Final Recommendation Table (one-glance)

Severity	Item	Effort	Surface
●	Move Anthropic key off the client	1 day	Results.tsx → Edge Function
●	Strip 14 production console.logs	30 min	Results.tsx, AssessmentEngine.tsx
●	Fix question-count copy inconsistency (35 vs. 56)	1 hour	Index.tsx, Dashboard.tsx, lib/config.ts
●	Single source of truth for STREAM_COLORS & careers	2 hours	lib/
●	Audit counsellor RLS post-migration-008	1 day	supabase/migrations/
●	Add ?stream= query param to Careers	1 hour	Careers.tsx
●	WhatsApp share deep-link	30 min	Results.tsx
●	Personalised reasoning instead of hardcoded paragraph	1 day	scoring.ts
●	Parent-specific narrative + parent chat	1 week	ParentDashboard.tsx + Edge Function
●	Subject-combination output (PCM/PCB/etc.)	3–5 days	scoring.ts + Results.tsx
●	Schools reference table + CSV bulk-import	1 week	migrations + CounsellorDashboard.tsx
●	Vitest + first tests on calculateScores	1 day	new
●	PostHog + Sentry	2 hours	App.tsx
●	Google OAuth + progressive profiling	1 day	Signup.tsx
●	Individual career pages /careers/:slug (SEO)	1 week	Careers.tsx
●	Cohort percentile view	2 days	Results.tsx
●	Hindi i18n	1 week	new
●	Adaptive item-bank (IRT-lite)	2–3 weeks	new
●	College + CUET shortlist (post-Class-12 upsell)	1 month	new
●	Question shuffling, A11Y pass, offline buffering, etc.	various	various

5. Verdict

The product **already does the hard parts well**: multi-role auth, RLS, resume-from-pause, attempt history, deltas, share tokens, India-specific exam roadmap. The remaining work is *cleanup, security, and parent surface*, not architectural rework.

If you fix the 6 ● items and ship the 5 highest-leverage ● items (WhatsApp share, personalised reasoning, parent narrative, subject combos, analytics), this product moves from *"impressive MVP"* to *"counsellor sells it into 10 schools next term"* in roughly 6 engineering weeks.

The single biggest under-exploited asset in the codebase is the **counsellor + school dashboard**: it's where the migration history shows the most pain, where the data model is weakest (free-text school name), and where the monetisation lives. Spend the next month investing there and the product changes shape from "free quiz" to "school SaaS".

Codebase commit: `fac0bf0` (plus this analysis at `ebece6b`). Each finding above can be traced to a specific file & line number in the repo — no recommendations are hypothetical.